

Routing, Band, Modulation and Spectrum Assignment with Dedicated Protection in Multiband- Elastic Optical Networks

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Abstract—Recently, several studies have aimed at utilizing multiband transmission technology to meet the increasing traffic growth rate primarily caused by the augmentation of the 5G and 6G services in telecom networks. Resource allocation schemes in Multiband-Elastic Optical Network (MB-EON) combine advantages of flexible-grid and wide spectrum band to allow more data transmission in the optical transport network. In this paper, we address the topic of band and spectral resource allocation together with protection in MB-EON. We analyze five approaches: Dedicated Protection with flexible resource allocation (DPP-RBMSA), Dedicated bands for protection: DBP- C_w-L_b and DBP- C_b-L_w , and Hybrid Dedicated bands for protection (*hyb*-DBP- C_w-L_b , and *hyb*-DBP- C_b-L_w). In our simulation-based analysis, we strive to find the best possible approach aiming at lower blocking and higher protection levels. We have used Bitrate Blocking Probability (BBP), Backup to Working Resources Ratio (BWRR) and Band Spectrum Utilization (BSU) as the performance indicators. Categorization of incoming requests into Protected Requests (PR), UnProtected Requests (UPR), and Blocked Requests (BR) to evaluate the protection level provided by the five approaches for considered network traffic load. We identify *hyb*-DBP- C_b-L_w and DPP-RBMSA as two approaches with the best blocking and protection performance, respectively, in simulated conditions.

I. INTRODUCTION

Global data traffic is increasing at an exponential rate, and according to a quarterly report [1], monthly data traffic has reached 157 EB in 2024. With the ever-increasing and varying traffic characteristics in the core network, it is a challenge for network operators to meet the upscaling diverse needs. Building upon the existing infrastructure and improving spectral efficiencies (flexible grids, and adaptive modulation formats) seem viable mid-term solutions. Recently, exploring the spatial aspect to increase the available capacity has been a popular approach, with options for multiple fibers, multimode/multicore fibers, and multi-band networks. Multiband-Elastic Optical Networks (MB-EON) is more lucrative than other solutions

This work has been funded by the Federal Ministry of Education and Research in Germany (BMBF) under the project HyperCORE (Grant ID 16KIS2100).

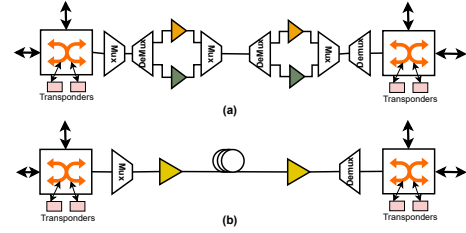


Fig. 1. Architectures to deploy C+L bands optical networks.

because unlike few-mode and multicore fibers, it does not require huge deployment investments and can be built over existing infrastructure in a brown-field manner.

MB-EON use Band Division Multiplexing (BDM) to extend the available bandwidth from C to L, S, E, and O bands. All of these bands together can provide 54 THz of capacity [2]. Commercialization of the S-, E-, and O-band components is still in progress, but deploying C+L MB-EON is already viable as L-band is operationally congruent with C-band Erbium Doped Fiber Amplifiers (EDFAs). A multiband network with C+L band operations will result in approximately 11.5 THz of bandwidth [3], and is expected to push the capacity limits for another few years. Different network upgrading approaches, [4]- [5] have been studied previously considering overall deployment costs, impairments consideration, and dispelling congested links. Two possible architectures for C+L deployments are: (i) two separate amplifiers (C and L EDFAs) in the link, requiring a large guard band, and (ii) a wideband amplifier, Fig. 1.

The excess capacity provided by the C+L with band-adaptive and bandwidth-variable technologies in resource allocation schemes enables MB-EON. The resources in MB-EON are Spectrum Slotss (SSs) in each band. Resource allocation schemes should have the best route selection (R), one of the two bands selection (B), appropriate Modulation Level (ML) selection as per route length-band characteristics, and SS reservation as per continuity and contiguity constraints. So, to accommodate every new connection request, an RBMSA is performed, and if resources/ SS are not available then the request is blocked.

Using the C+L band solutions network operators strive to push down the blocking of the connection requests, leading to the acceptance of more connection requests.

Ensuring acceptance of more traffic volumes together with resilience is another challenge for operators. Excess capacity in MB-EON will surely allow more traffic volume in the network, and any type of failure can result in significant losses for network operators. The types of failures that can occur in MB-EON are- node failures or link failures (fiber cut or amplifier failure). Network resilience can be ensured by either protection or restoration schemes. In protection schemes, redundant resources are reserved in advance for quick failure responses with instant switchover from working to backup resources. Restoration schemes find backup resources once the failure occurs, and therefore cannot ensure 100% resilience. Due to complex provisioning operations involving two bands in C+L MB-EON, 100% survivability is achievable with proper protection approaches. Protection can be provided with dedicated or shared resources. Dedicated Path Protection (DPP) requires duplicated resources (1+1) through two different end-to-end paths/routes. Switching to backup at the receiver takes place in the event of failure. There are hot and cold backup strategies, based on whether backup resources will always be active or activated only after failure. Although shared path/link protection ensures efficient utilization of the resources than DPP, it also requires plenty of provisioning constraints like finding shared risk link groups and longer recovery time.

In this paper, we have proposed and compared approaches to provide resilience in MB-EON using DPP with flexible and dedicated band options. To relax the computation complexity of the flexible approach, we have proposed Dedicated Band Protection (DBP), for working and protection resources limited to either C- or L-band. Also, a hybrid DBP, where only backup resources are limited to either C- or L-band, whereas working resources may be in either band. In this paper, we present:

- i. DPP with the flexible band and SS selection approach in MB-EON scenario with wideband amplifiers architecture, where it is mandatory to have link-disjoint paths for primary and backup resources
- ii. DBP + DPP and *hyb*-DBP + DPP as alternative approaches to relax the computation complexity and utilize bands' spectral resources efficiently
- iii. analysis of the trade-offs between protection levels and blocking across the proposed schemes.

In Section II, relevant literature from other research works is covered. Section III presents the overview of the proposed techniques with examples. Section IV has a detailed analysis of the network performance. Section V will provide our derived conclusions.

II. RELATED WORKS

Survivability in MB-EON has been reported in numerous works. Authors in [6], presented survivable resource

provisioning in a multicore and multiband-enabled core optical network using a band partition and cold backup approach, and assessed the approaches for blocking, fragmentation, impairments, and protection level. Authors in [7] have reported GSNR-sensitive flexible survivability schemes in multiband network scenarios. This paper covers the impact of link failures on the neighboring channels, and the requirements for cost-effective resiliency using a shared restoration mechanism. Authors in [8] proposed implementing multi-band transmission in only a few links and that too as a virtual bypass, to be shared by backup paths. Authors in [2] have demonstrated dynamic restoration assessment for optical amplifier failures in partially upgraded multi-band networks with separate amplifier architecture and compared the performance with different DPP schemes. Authors in [9] have proposed a hybrid resource allocation approach to provide at least protection against amplifier failure. Authors in [10] have also presented dedicated protection against fiber and amplifier failures, where L-band has primary resources and its hybrid version, concluding that there is a trade-off between protection and blocking. Authors in [3] have proposed a differentiated protection mechanism based on Service Level Agreement (SLA) requirements while using DPP and Shared Backup Path Protection (SBPP). Many approaches in the above works focus on improving the blocked requests number, and resilience against amplifier failures only. In this paper, we focus on the resource provisioning scheme with dedicated path protection, considering networks with fully upgraded links and the usage of wideband amplifiers, which strictly requires link-disjoint routes covering both fiber cut and amplifier failures.

III. PROPOSED METHODOLOGY

The proposed protection approaches are explained in the following section for MB-EON represented as a graph $G(N, L)$, with N number of nodes and L number of links. Each Single Mode Fiber (SMF) link has two operating bands C and L, comprising spectrum slots SS_C and SS_L , respectively. When a connection request CR arrives between source 's' and destination 'd', with a bandwidth requirement of BW , there is a need to perform RBMSA to find the available resources, and to admit the CR . The K-shortest (link-disjoint in protected RBMSA) paths are pre-calculated for all $s - d$ pairs. On the selected route, modulation level m is chosen based on the bands' transmission reach table. The required number of spectrum slots, SS_r are calculated by using the formulation, $SS_r = \lceil \frac{BW_r}{2 * CW * m} \rceil + 1$. CW is the spectrum slot width and we have considered it as 12.5 GHz. One additional spectrum slot is added as the guard band. Now, these many SS_r are looked for in the links of the selected route following the spectrum continuity, contiguity, and non-overlap constraints. If resources are available, then the CR is accommodated, otherwise rejected.

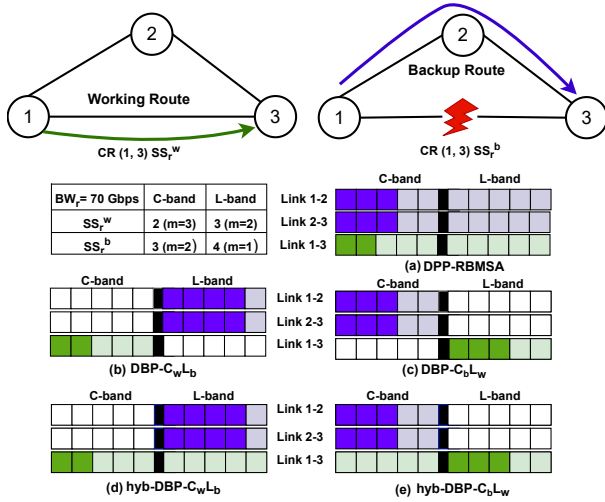


Fig. 2. Given 3-Nodes 3-Links network, with working and backup routes. If there are two bands and three modulation levels available ($m = 1, 2, 3$ for BPSK, QPSK, 8-QAM), then table shows SS_r^w and SS_r^b for bands and route combinations. Since, L-band is less robust than C-band, leading to additional SSs. Proposed Approaches with link-disjoint routes for working and backup resources, where (a) DPP-RBMSA has search space of both C-and L-band for working (in green color) and backup (in purple color) with first preference to C, (b) DBP-C_w-L_b has working resources strictly in C-band and backup resources strictly in L-band, (c) DBP-C_b-L_w has working resources strictly in L-band and backup resources strictly in C-band, (d) *hyb*-DBP-C_w-L_b has only backup resources strictly in L-band, and (e) *hyb*-DBP-C_b-L_w has backup resources strictly in C-band

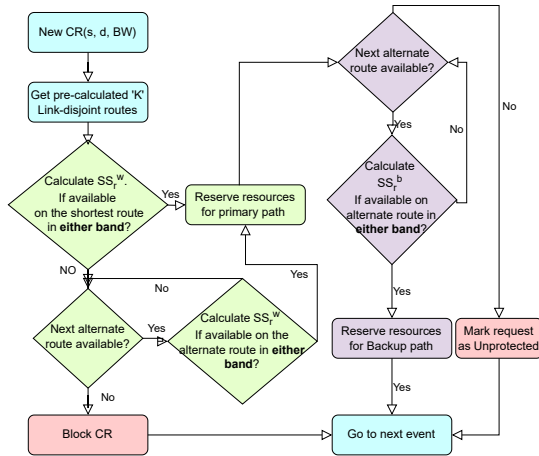


Fig. 3. Flowchart for DPP-RBMSA.

The survivability aspect is ensured by duplicating resources for a CR on the available resources. In the DPP scheme, working and backup routes are established, and both carry traffic end-to-end, but the receiver side decides the active connection based on the signal quality. The backup route needs to be link-disjoint from the working route to ensure full protection for the CR . The required resources for the two link-disjoint routes are, SS_r^w for working route and SS_r^b for the backup route. The link-disjoint backup paths are usually longer than the working

paths, and this leads to $SS_r^w \leq SS_r^b$. In the MB-EON, the available resource pool is from two bands, and the transmission reach characteristics are different for them, as explained in [11], using a loss model. C-band is more robust than L-band, that is C-band has longer transmission reach than L-band for same modulation level in C+L MB-EON with BER of 10^{-6} . In our proposed approaches, we leverage these reach and band robustness characteristics to utilize the available resources most efficiently. The objective is to have lower incoming request blocking, with a higher level of protection.

Fig. 2 shows a 3-nodes and 3-links network with a $CR(1, 3)$ with bitrate requirement of 70 Gbps. The routes, available m levels and different SS requirements for working and backup are shown in the table in Fig. 2. The five proposed approaches addressing the resource allocation are as follows:

DPP-RBMSA: It is a flexible and first-fit approach for the routes, band selection, and spectrum slots. The search space for working and backup resources includes both bands, but requires link-disjoint paths, to ensure protection against both fiber and amplifier failures. When a connection request CR arrives, the set of link-disjoint routes is retrieved from the pre-computed routes. First, we try to establish the working slots on the shortest route. A first preferred band (i.e., C) is selected, and based on the length of the route, the most robust modulation level is selected, and the corresponding SS_r^w is calculated. If SS_r^w are available then along the links of the selected route the CR is established. If not, the next band is checked, and if still not available then the next route is checked repeating the previous steps, as shown in Fig. 3. If resources are not available in any of the bands and routes, then CR is blocked. SS_r^w for CR are reserved, once found, and at this stage, the CR is accepted with the status 'Unprotected'. If there is any alternate route available, then the search for the backup resources is performed with SS_r^b calculated corresponding to the route length and band, first in C-band and then L-band, Fig. 3(a). If resources are available then the CR status is updated to 'Protected', and if not, then the RBMSA process is completed with CR being accepted but 'Unprotected'.

DBP-C_w-L_b: In this dedicated band approach, C-band is strictly used for provisioning working spectral slots and the L-band is strictly used for backup spectral slots. The search space is limited to designated bands as shown in Fig. 2(b). The use of the C-band for working routes may result in more efficient utilization of the available resources, as the transmission reach is more than the L-band for different modulation formats. This technique may result in higher utilization of resources in both bands, in the C-band due to more acceptance of CR s and in the L-band due to longer path lengths- m combination and hence, SS_r^b .

DBP-C_b-L_w: This is again a dedicated bands and paths protection approach, where the L-band is used for

provisioning working resources and the C-band is used for backup resources. Again the search space is limited to its designated bands as shown in Fig. 2(c). The use of the L-band for working routes may result in more utilization of the available resources, but more acceptance of CRs is expected due to extra capacity available in comparison with the C-band. In this approach, C-band is expected to have high utilization than in DBP- C_w - L_b but fewer protected requests.

hyb-DBP- C_w - L_b : In this hybrid approach, there is an additional step than DBP- C_w - L_b , which is to find working resources in the L-band as well if not available in C-band. The search space for the working resources (in green color) is extended in L-band, Fig. 2(d). This may lead to more acceptance of CRs and more utilization of resources in L-band than DBP- C_w - L_b , but simultaneously limiting the number of protected requests, because of accommodation of CR even without backup is allowed.

hyb-DBP- C_b - L_w : This hybrid approach extends the search space of finding working resources in the C-band if not available in the L-band. The search space for the working resources (in green color) is extended in C-band, Fig. 2(e). This may again lead to more acceptance of CRs and more utilization of resources in L-band than DBP- C_b - L_w but at the cost of protection level.

In this paper, we have considered a full C+L multiband network, with wideband amplifier architecture, and compared the performance of the above mentioned approaches in several nationwide core network topologies. Due to differences in the transmission reach tolerances of the two bands, and varying requirement of the working and backup paths, we expect to see a trade-off between the accepted CRs , blocked bitrate and protection levels in the five approaches.

IV. PERFORMANCE EVALUATION

We compare the impact of the protection provisioning in six network topologies listed in Table I. We assume that network links are C+L capable, and have 344 C-band and 480 L-band SSs respectively [12]. The spectrum slot width of 12.5 GHz is considered. We have used an event-driven Matlab simulator, where arriving CRs are generated using Poisson distribution with an average arrival rate of λ per node $\{40, 90\}$ for small-scale and $\{15, 40\}$ for large-scale networks. Average holding time of 1 unit, i.e., μ , is considered with exponential distribution. Network traffic load in plots is $\frac{\lambda}{\mu} \times \text{Total nodes}$. The source and destination of CRs are selected randomly following a uniform distribution. CRs may have bitrate demands of 100, 200, and 400 Gbps, again selected uniformly. $K=3$ link-disjoint paths are pre-calculated for all the $s-d$ pairs. Four modulation levels: DP-BPSK, DP-QPSK, DP-16-QAM, and DP-64-QAM are considered with a BER threshold of 10^{-6} [11]. First-fit spectrum allocation is used in all the approaches. CRs are accommodated if the spectrum assignment constraints are satisfied. After arrival and

departure of 50k CRs , the network simulator reaches a steady state, and this decided based on transient analysis of all the considered topologies. After that, performance indicators are recorded for next 100k arrived CRs in the network.

TABLE I
NETWORK PARAMETERS

Network	Nodes	Links	Avg. Degree	$K=1 P_l$ (km)
US-24	24	42	3.5	6650
Panero	28	39	2.78	5101
Coronet	75	99	2.64	6472
Cost239	11	26	4.73	1660
DT-17	17	26	3.06	951
JP-12	12	17	2.08	2963

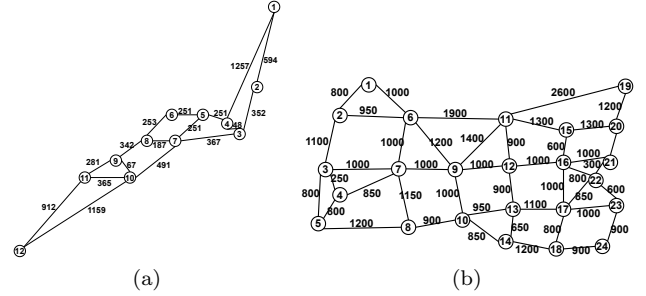


Fig. 4. Evaluated Network topologies: JP-12 and US-24

Out of the six networks, we present the performance plots of the JP-12 and US-24 networks, Fig. 4. We chose these topologies because they have diverse network characteristics in terms of size and average node degree. The key performance indicators to evaluate the proposed approaches are:

1) *Bitrate Blocking Probability (BBP)*: Ratio of blocked bitrate to requested bitrate. A lower BBP value indicates better performance. BBP vs network traffic load plots for JP-12 and US-24 are shown in Fig. 5a and Fig. 5c, respectively.

2) *Backup to Working Resources Ratio (BWRR)*: Average ratio of $(SS_r^b \times \text{Hops(Backup - route)})$ and $(SS_r^w \times \text{Hops(Working - route)})$ for protected requests. It indicates the relative redundancy required to provide resilience to protected requests. BWRR vs network traffic load plots for JP-12 and US-24 are shown in Fig. 5b and Fig. 5d.

3) *Band Spectrum Utilization (BSU) of C and L bands*: Ratio of used SSs to total SSs in individual bands in the network. A higher utilization with lower BBP is desired, as it indicates the available resources are being used efficiently, and there are sufficient resources for future CRs . BSU for C and L vs highest network traffic load plots for JP-12 and US-24 are shown in Fig. 6a and Fig. 6b, respectively.

4) *Indexing of connection requests*: Protected (PR), Unprotected (UPR), and Blocked (BR). In the best-case scenario, we expect more PRs, fewer UPRs, and no BRs. Total connection requests distribution among PR, UPR, and BR at the highest traffic load for respective networks for all six networks are tracked in Table II.

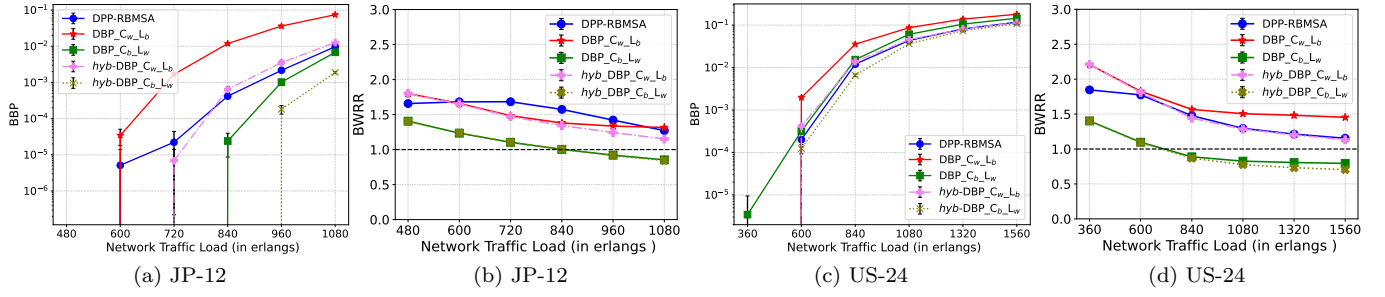


Fig. 5. BBP and BWRR performance indicators for JP-12 and US-24

In the JP-12 network, when comparing fixed DBP approaches, DBP- C_b - L_w has lower BBP than DBP- C_w - L_b , as shown in Fig. 5a. In *hyb*-DBP- C_b - L_w , there is flexibility to search working resources in the backup band ‘C’ as well, and BBP is further reduced. Other flexible approach *hyb*-DBP- C_w - L_b also has better BBP performance than fixed DBP approaches. Lower BBP in the hybrid approaches comes at the cost of the 1-2 % of reduced protection level, Table II. At highest considered traffic load, *hyb*-DBP/DBP- C_w - L_b has 7-8% more protection than *hyb*-DBP/DBP- C_b - L_w , because of more L-band capacity than in the C-band, and hence the capability to fulfill increased SS_r^b requirements. There is reduced BR% in *hyb*-DBP- C_w - L_b than DBP- C_w - L_b because of the more available capacity in L-band for SS_r^w . It is shown in Fig. 6a that *hyb*-DBP/DBP- C_w - L_b uses more resources in L-band than *hyb*-DBP/DBP- C_b - L_w . If L-band acts as backup then there is more SS_r^b requirement, leading to more L-band BSU. DPP-RBMSA utilizes mostly C-band resources, as there is first-fit allocation, starting with C-band, for both working and backup routes, and hence C-band BSU is higher. DPP-RBMSA also gets less blocking than the DBP- C_w - L_b and *hyb*-DBP- C_w - L_b , as it is trying to utilize the available resources as they are available, rather than limiting the SS_r^w to C-band only. DPP-RBMSA has fairly more PR% than other techniques, shown in Table II. BWRR gives insight into how much excess resources are used by the PRs. C_b - L_w , both fixed and hybrid, have BWRR closer to one, which means $SS_r^b = SS_r^w$ owing to the C-band’s better transmission reach tolerances than L-band. *hyb*-DBP- C_b - L_w has least BBP, most efficient BWRR but with lowest PR% among all approaches. Performance summary of five approaches for JP-12 is shown in Table II. DPP-RBMSA has significantly higher PR% than *hyb*-DBP- C_b - L_w and DBP- C_b - L_w . From the figures and Table II, we see that BBP improvement of 80% in *hyb*-DBP- C_b - L_w over DPP-RBMSA comes at the cost of loss of 11.48% protection. In small-scale networks, the (*hyb*)-DBP with L as a working band and C as a backup band could be a better provisioning approach with 80% less BBP than DPP-RBMSA and 79.9% of PRs at highest traffic load value. In Table II, we see that DT-17, which is another small-scale network has similar performance as JP-12. In

DT-17, *hyb*-DBP- C_b - L_w outperforms other approaches in terms of BR% but has considerably lower PR% than other techniques. Cost239, which is a well-connected network with avg. node degree of 4.7, shows no blocking and maximum protection for the hybrid approaches.

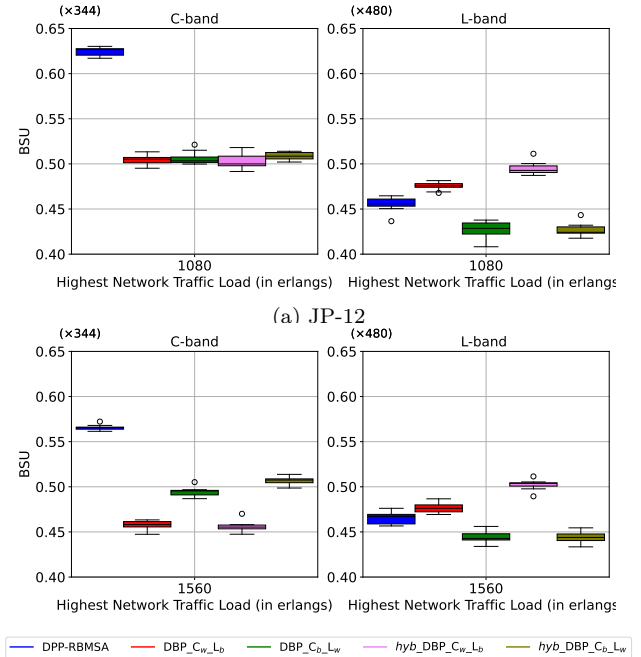


Fig. 6. BSU at the highest traffic load for JP-12 and US-24 networks.

For a geographically vast network like US-24, in Fig. 5c again *hyb*-DBP- C_b - L_w has lower BBP than other approaches followed closely by DPP-RBMSA having 1.6% more blocked bitrate. We also find that DPP-RBMSA has the highest PR%. With DPP-RBMSA BBP close to *hyb*-DBP- C_b - L_w BBP, there are still 7% more PRs, as shown in Table II. PR% is least for *hyb*-DBP- C_b - L_w among other approaches, and this leads to an observation that more CR acceptance comes at the cost of the protection. The fixed DBP approaches do not perform well in terms of BBP in this network, with 4% more blocking than hybrid approaches. *hyb*-DBP- C_b - L_w and DBP- C_b - L_w has comparatively least redundancy in terms of BWRR value, but their counterpart *hyb*-DBP- C_w - L_b and DBP- C_w - L_b have the worst backup redundancy because of longer routes

TABLE II
PERCENTAGE OF PROTECTED, UNPROTECTED AND BLOCKED REQUESTS AT HIGHEST TRAFFIC LOAD CONDITIONS IN TEST NETWORKS

Test Network	DPP-RBMSA			DBP- C_w - L_b			DBP- C_b - L_w			hyb-DBP- C_w - L_b			hyb-DBP- C_b - L_w		
	PR	UPR	BR	PR	UPR	BR	PR	UPR	BR	PR	UPR	BR	PR	UPR	BR
US-24	79.42	13.32	7.24	77.20	11.03	11.75	74.16	16.56	9.269	74.84	17.99	7.15	72.38	21.01	6.61
Paneuro	70.6	22.11	7.29	68.6	17.96	13.37	64.03	27.32	8.64	65.67	26.86	7.46	62.37	31.67	5.95
Coronet	60.89	23.93	15.18	61.23	17.48	21.28	57.55	25.18	17.26	57.57	27.72	14.71	55.17	31.06	13.75
Cost239	83.48	12.48	4.04	83.86	7.30	8.83	75.21	22.56	2.23	99.89	0.11	0	96.27	3.73	0
DT-17	91.9	7.74	0.34	91.91	4.01	4.06	84.09	15.83	0.07	90.82	8.68	0.491	84.19	15.83	0.02
JP-12	90.91	8.51	0.57	88.88	6.67	4.44	80.11	19.49	0.40	87.62	11.64	0.74	79.90	19.99	0.11
Best :)			Neutral :	Worst :(			Neutral :			Worst :(			Best :)		
■ Best Protection%			■ Least Blocking%	■ Worst Blocking%			■ Worst Protection%			■ Second Best BR%					

requiring more SS_r^b in L-band, Fig. 5d, also reflected in BSU plots in Fig. 6b. From Table II we can assert that in large-scale networks, DPP-RBMSA has 0.63% more blocking but 7-8% more PRs than blocking-wise best performing *hyb*-DBP- C_b - L_w . Similar to US-24, for Paneuro and Coronet DPP-RBMSA appears a better approach because, for a trade-off of 1-2% of more blocking it provides 3-8% of more protected requests than hybrid approaches. Looking into PR% and BR% of all large-scale networks, we conclude that flexible approaches emerge as clear winners, with DPP-RBMSA having the best performance scenario because protection% improvement is more significant than difference in excess blocking over hybrid approaches.

V. CONCLUSION

In this paper, we evaluated the performance of Dedicated Path Protection with/without Dedicated Bands in a C+L MB-EON Scenario. We have a flexible DPP-RBMSA with route, band, and spectrum allocation as per availability, for both working and backup resources. Other approaches have Dedicated Bands for working and backup resources, DBP- C_w - L_b , and DBP- C_b - L_w . We also evaluated the hybrid version of DBP approaches, *hyb*-DBP- C_w - L_b , and *hyb*-DBP- C_b - L_w , where only backup resources are limited to one band while working resources have the search space of both bands. Our simulation-based analysis leads us to believe that in small-scale networks like JP-12, the DBP approach with L-band as working and C-band as backup performs better in terms of lower bitrate blocking and BWRP closer to one, but it comes with a penalty on the protection percentage. There is a further lowering in BBP if the search space of working resources is extended to both bands in *hyb*-DBP- C_b - L_w . In an extensive network like US-24, *hyb*-DBP- C_b - L_w still has the lowest BBP closely followed by DPP-RBMSA, but the protection level in DPP-RBMSA is significantly more than the *hyb*-DBP- C_b - L_w . In such networks, DPP-RBMSA could be preferred over the *hyb*-DBP approaches, with an insignificant trade-off in blocking. In the proposed approaches, the configuration selection scheme is only modulation-transmission reach threshold based but the assessment of the impact of other configurations and sharability of backup resources on the accommodation of CRs need to be investigated.

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